

Juan Glicerio C. Manlapaz

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PROFESSIONAL PROFILE

- A registered electronics engineer and electronics technician with three years of experience in power electronics and research. Specialized in DC/DC converter design with a proven track record of research presentation and publications. Recognized as a highly versatile and self motivated individual with technical skills ranging from programming to simulation.

EDUCATION

University of the Philippines Diliman

September 2023 – June 2026

M.S. in Electrical Engineering

- MS Thesis: *Hybrid magnitude control in LLC resonant converters for fuel cell applications*
- Adviser: *Carl Michael Odulio, PhD*
- Grade: *1.46/1.00*

Ateneo de Manila University

August 2018 – December 2022

B.S. in Electronics Engineering with Minor in Data Science and Analytics

- Undergraduate Thesis: *Optimization and Simulation of Distributed Generation: A Case Study on the PLDT-CTC Building of Ateneo de Manila University*
- Distinction: *Cum Laude*
- Grade: *3.63/4.00*

CERTIFICATIONS

Registered Electronics Engineer, PRC ID No. 0079282

Registered Electronics Technician, PRC ID No. 0026363

WORK EXPERIENCE

Senior Lecturer I

August 2026 – Present

University of the Philippines Diliman

Quezon City

Power Electronics Engineer

February 2026 – Present

Nascent Technologies Corp.

Quezon City

- Based in Metro Manila, Philippines, Nascent Technologies aspires to be Southeast Asia's leading provider of battery energy storage technology and solutions for stationary and transportation applications.
- The primary responsibility of a Power Electronics Engineer is the design, fabrication, and deployment of power electronics for energy storage products. This includes the development of power conversion and control systems that enable efficient energy storage and integration with the electrical grid and off-grid systems.

Graduate Assistant

September 2025 – July 2026

University of the Philippines Diliman

Quezon City

- Responsible for creating power electronics laboratory activities involving Analog Devices tools for undergraduate and graduate students.

ERDT Student (MS)

August 2024 – June 2026

University of the Philippines Diliman

Quezon City

- MS EE student under the Engineering Research and Development for Technology (ERDT) Scholarship Program.
- Currently conducting power electronics research in the Artesyn-Power Electronics Laboratory, Electrical and Electronics Engineering Institute. MS thesis focuses on *LLC* resonant converters for fuel cell applications.

Research Fellow I (Power Electronics), Project CIPHER

June 2023 – August 2024

University of the Philippines Diliman

Quezon City

- Project CIPHER (short for Clean and Vertically-Integrated Pure/Applied Hydrogen Energy Research for Next Generation Power Systems) aims to develop energy storage technologies by addressing key areas in energy research, combining computational and experimental approaches to help the Philippines transition into a cleaner and sustainable energy system.
- Funded by the Philippine California Advanced Research Institutes (PCARI) program of the Commission on Higher Education (CHED) and is currently under the joint supervision of Laboratory of Electrochemical Engineering and Artesyn-Power Electronics Laboratory, University of the Philippines Diliman.
- Implemented power electronics projects such as DC/DC converters for fuel cell systems and presented the findings in various scientific conferences.

Intern, CIO - Travel and People Mobility - IT

July 2021 – August 2021

Accenture in the Philippines

Mandaluyong City

- On-the-job Training
- Developed automation code for addressing developer backlogs, resulting in a 25% increase in bug processing
- Developed 3 automation scripts for addressing bugs in a span of two weeks.

PROFESSIONAL MEMBERSHIPS

Institute of Electrical and Electronics Engineers (IEEE)	2025 – Present
Institute of Electronics Engineers of the Philippines (IECEP)	2023 – Present

PUBLICATIONS

J. G. Manlapaz and C. M. Odulio, "Hybrid magnitude control in LLC resonant converters utilizing phase shift and variable frequency control," *2026 4th IEEE International Power Electronics and Application Conference (PEAC)* (**Accepted**)

J. G. Manlapaz and C. M. Odulio, "A resonant frequency tracking algorithm for LLC resonant converters based on primary voltage waveforms," *2026 4th IEEE International Power Electronics and Application Conference (PEAC)* (**Accepted**)

J. G. Manlapaz, A. Escultero and C. M. Odulio, "Accuracy Analysis of the Fundamental Harmonic Approximation for LLC Resonant Converters," *2025 IEEE 10th Southern Power Electronics Conference (SPEC)*, Johannesburg, South Africa, 2025, pp. 1-6, doi: 10.1109/SPEC64875.2025.11377205.

J.G. Manlapaz, D. Reyes, C. P. Buensuceso, R. A. Peña, R. Parocha, and E. Q. Macabebe, "Optimization and simulation of a grid-connected PV system using load forecasting methods: A case study of a university building", *IOP Conference Series: Earth and Environmental Science*, 2023, vol. 1199, doi: 10.1088/1755-1315/1199/1/012006

C. P. L. Buensuceso, **J. G. C. Manlapaz**, D. M. Reyes, R. A. S. Peña, R. C. Parocha and E. Q. B. Macabebe, "Energy Management for Grid- Tied PV Systems Using Particle Swarm Optimization," *2022 IEEE International Conference on Power and Energy (PECon)*, 2022, pp. 105-110, doi: 10.1109/PECon54459.2022.9988838

CONFERENCES

J.G. Manlapaz, A. Escultero, and C. M. Odulio, "Accuracy Analysis of the Fundamental Harmonic Approximation for LLC Resonant Converters", 10th Southern Power Electronics Conference, 2025

J.G. Manlapaz, A. Escultero, O. Salvaleon, J. Ocon, and C. M. Odulio, "Performance Evaluation of LLC Resonant Converter Configurations for Fuel Cell Applications", 12th Engineering Research and Development for Technology Congress, 2024

J.G. Manlapaz, A. Escultero, J. Ocon, and C. M. Odulio, "Optimal Sizing of Hybrid PEMFC/Lithium-Ion Battery Power Systems Using Rule-Based Cycle Charging Algorithm for Telecommunication Base Stations", 2024 National Electrical, Electronics, and Computer Engineering Conference, 2024

J.G. Manlapaz, A. Escultero, O. Salvaleon, J. Ocon, and C. M. Odulio, "Comprehensive Modeling of Internal Resistance in Lithium-Ion Batteries Across the Full State of Charge Range", 2024 Annual PAASE Meeting & Symposium, 2024

J.G. Manlapaz, A. Escultero, O. Salvaleon, J. Ocon, and C. M. Odulio, "Performance Evaluation of LLC Resonant Converter Configurations for Fuel Cell Applications", 2023 International Conference on Sustainable Energy and Green Technology, 2023

C. P. Buensuceso, **J.G. Manlapaz** , D. Reyes, R. A. Peña, R. Parocha, and E. Q. Macabebe, "Energy Management System for a University Building Microgrid using Particle Swarm Optimization", 2022 International Conference on Power and Energy, 2022

J.G. Manlapaz, D. Reyes, C. P. Buensuceso, R. A. Peña, R. Parocha, and E. Q. Macabebe, "Optimization and simulation of a grid-connected PV system using load forecasting methods: A case study of a university building", 2nd ASEAN International Conference on Energy and Environment, 2022, *Best Presenter*

SPEAKING ENGAGEMENTS

Invited Speaker <i>Junior High School Department, Columban College - Barretto</i>	10/14/2024
Gave a talk entitled "Strand Orientation for Grade 10 - STEM" to Grade 10 students	
Invited Speaker <i>College of Engineering, Columban College</i>	04/19/2024
Gave a talk entitled "Exploring Future Frontiers: Advancements and Innovations in Electrical, and Computer Engineering" to engineering students	

VOLUNTEERING WORK

Office of Admission and Aid, Loyola Schools, Ateneo de Manila University <i>Scholar Volunteer</i>	2018 – 2022
- As a financial aid scholar in Ateneo de Manila University, worked as one of the student volunteers of the Office of Admission and Aid (OAA) as part of service hours requirement among others. - Demonstrated versatility, competence, compassion, and efficiency in various roles asked by the office. - Roles include Ateneo College Entrance Exam (ACET) OAA Assistant and Proctor, Campus Tour Guide, Infodesk Volunteer, Data Encoder, Video Editor, and Academic Tutor.	
Ateneo Electronics and Computer Engineering Society <i>Executive Board Member (2020 – 2021), Central Board Member (2019 – 2020)</i>	2019 – 2021
- Organized 7 technical workshops ranging from Basic Arduino to Git Fundamentals. - Organized departmental events and schedules which increased workshops offered by 100% and tutorial services given by 200%. - Compiled 25 electronics and computer engineering academic resources into a centralized folder aimed at assisting 160 electronics engineering and computer engineering majors in understanding the new curriculum. - As an overseer, worked as the main externals head in communicating prospective speakers for the event, led a team of freshmen in organizing the event.	
Learning Management System Support Services <i>Member</i>	2020
- A project under a team led by the Department of Information Systems and Computer Science (DISCS) professors. - Trained and handled ten professors from the Loyola School of Theology, School of Social Sciences, and John Gokongwei School of Management to be familiar and competent with their respective LMS to facilitate them in their transition to online instruction during the COVID-19 pandemic.	

SKILLS

Programming Languages: C++, Python, SQL (MySQL), Arduino, Julia, HTML, CSS, Bootstrap, MATLAB, Simulink
Libraries: pandas, NumPy, Matplotlib, SymPy
Miscellaneous Software: Notion, LaTeX, AWS, Tableau, LTSpice, NI MultiSim, HOMER Pro, Adobe Creative Cloud, CAD, KiCAD, JMP, MPLAB X IDE, SIMetrix, PLECS

AWARDS AND HONORS

ERDT Scholarship	2024 - 2026
Financial Aid Scholarship	2018 - 2022
DOST-SEI Merit Scholarship	2018 - 2022